Quantified Self

1 Book:Machine Learning for the Quantified Self

Reference [HF17]: 239 pages. Read up to page: 27.

Definition 1: Quantified self: an individual engaged in self-tracking of any kind of biolgical, behavioral information. There is a proactive stance toward obtaining information andacting on it.
Definition 2: the quantified self is any individual engaged in self-tracking of any kind of biological, physical behavioral or environmental information. The self-tracking is driven by a certain goal of the individual with a desire to act upon the collected information.
Three purposes: improve health, enhance aspects of life, find new life experiences. Other survey cites: self-healing, self-discipline, self-design, self-association (being part of a community and relate yourself to the community).
Self: healing, discipline, design, association, entertainment = the five factor framework of ST motivations.
Machine learning applied to quantified self-data: extract patterns from collected data and enable a user to act upo insights effectively, which contributes to the goal of the user..
Unique characteristics of machine learning in the quantified self context:

- sensory data is noisy
- measurements are sometimes missing
- data has a highly temporal nature
- algorithms should enable support and interaction with users without a long learning period
- we collect multiple datasets and can learn across them

1.1 Terminology

1.1.1 Data Terminology

- **Attributes:** also called features or variables.
- **Measurement**: a value for an attribute recorded at a specific time point. Measurements can be numerical or categorical, with an ordering (ordinal) or not (nominal). The time point is not part of the attributes, it is an inherent part of the measurement itself. Measurements often come in sequences: they are called time series.
- **Time series**: a series of measurements in temporal order.

1.1.2 Machine Learning Terminology

- Supervised Learning: the ML task of inferring a function from labeled training data.
- Unsupervised learning: example clustering and outlier detection. In unsupervised learning there is no desired outcome.
- Semi-supervised learning: ML techniques that learn pattern in the form of a function based on labeled and unlabeled training data
- Reinforcement Learning: find the optimal actions in a given situation, to maximize a numerical reward that does not immediately come with the action but later in time. In RL, the learner learns which actions yield the highest reward over time, by trying them,

1.1.3 Mathematical notation and formulas

Input variables are denoted by X. X can be a vector., containing multiple variables, i.e. multiple attributes or features. We assume that every X has p variables or attributes. For example each X is composed of X_1 = heart rate, X_2 = speed.
In case of supervised learning, the output are denoted as:

- Y in regression
- G in classification

Again, the subscript is used to indicate one of the multiple outputs.
An observation of X is a single instance of the data, with observed values for all variables) is denoted as x_j : it is a column vector of observation of p variables. $j \in [1, N]$, where N is the number of observations.
To refer to the value of a variable within an observation the notation is $x^k{}_j$, where $k \in [1, p]$
We can represent the entire dataset as a matrix $\boldsymbol{X}$ (bold X). The dimensions are $N \times p$, so N rows and p columns.
$x_{i,j}$ = the jth variable of the ith instance. i Is the first index, it represents the sample.
$\boldsymbol{X}_i$ (bold X): denotes a specific dataset, originating from a person, for example. $\boldsymbol{X}^{\boldsymbol{\tau}}$ is a time series, where instances are ordered. $\boldsymbol{X}$ Does not amek assumptions about the ordering.
Notation for the output. Predicted output is $\hat{\boldsymbol{Y}}$ or $\hat{\boldsymbol{G}}$. Ground truth is $\boldsymbol{Y}$ or $\boldsymbol{G}$. y_i and g_i :individual instances

2 Glossary of acronyms and terms

Term	Meaning
Machine Learning	Automatically identifying patterns from data
QS	Quantified Self
ST	Self Tracking

3 Biography

ID	Title	Version
[HF17]	"Machine Learning for the Quantified Self", M. Hoogendoorn,B. Funk, Springer, 239 pages https://ml4qs.org/ : Machine learning for the Quantified Self: website.	2017